

COMPLEX ENGINEERING PROBLEM (CEP)

COURSE TITLE: Computer Networks
INSTRUCTOR: Dr. Salman
Lab Engineer: Engr. Usama Shoukat
COURSE CODE: EC330
SEMESTER: 5th (Fall-2024)
ENTRY: DE44CSE
Credit Hours: 3(Theory)+1(Lab)

Internet Protocol version 6 (IPv6) is the most recent version of the Internet Protocol (IP), the communications protocol that provides an identification and location system for computers on networks and routes traffic across the Internet. With the rapid growth of the Internet after commercialization in the 1990s, it became evident that far more addresses would be needed to connect devices than the IPv4 address space had available. By 1998, the Internet Engineering Task Force (IETF) had formalized the successor protocol. IPv6 uses a 128-bit address, theoretically allowing 2^{128} , or approximately 3.4×10^{38} addresses. The total number of possible IPv6 addresses is more than 7.9×10^{28} times as many as IPv4, which uses 32-bit addresses. Therefore, the goal is to implement ipv6 addressing instead of ipv4 addressing in our networks.

CEP Statement:

Design an IPv6 based WAN (wide area network) using available hardware in the lab (Cisco Router, switches).

In the planning stage, select the network devices and justify your selection. In the implementation stage specify the parameters of each device.

Implement ipv6 addressing. Connect minimum of three 3 end devices. All devices and active ports shall ping each other. You should give proper reasoning of protocols and basic configurations followed in the network.

Course	WK	PLO (WA)	WP	Bloom's Learning Level
Computer Networks	WK3, WK4	3 (Design)	WP1, WP2, WP3, WP4, EP2	C6

Deliverables:

- Complete design of the network topology
- Configuration commands
- Project report addressing all requirements

Pre-Requisite:

To carry out this complex engineering problem you should have the knowledge and understanding of

- Networking devices
- Routing protocols
- IP addressing and subnetting.